

Notes: Giroud & Mueller (JFE, 2010)

Does corporate governance matter in competitive industries?

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1 Executive overview

1.1 The core question

Corporate governance is often studied through the lens of *external discipline*, especially the threat of takeovers. This paper asks a question:

Does external corporate governance matter equally across product markets, or is it largely redundant when product market competition is intense?

1.2 The central hypothesis

Product market competition can act as a substitute disciplining device:

- In **competitive industries**, managers have limited room to slack off because poor performance quickly threatens survival.
- In **non-competitive industries**, managers can enjoy more slack; thus changes in takeover pressure should affect real decisions and performance more strongly.

1.3 The quasi-experimental shock

The paper uses the staggered adoption of **business combination (BC) laws** across U.S. states (1980s–early 1990s). BC laws are anti-takeover statutes that make hostile takeovers harder, thereby plausibly *weakening external governance*.

1.4 Main result

After BC laws weaken takeover pressure, **operating performance (ROA) declines in non-competitive industries but is largely unchanged in competitive industries**, implying that product market competition and external governance are substitutes.

2 Data and sample construction

2.1 Sample and period

- Main data: Compustat (accounting), CRSP (returns used for event study).
- Sample: U.S. firms *located and incorporated* in the United States; exclude regulated utilities (SIC 4900–4999); exclude missing/negative assets or sales.
- Sample period: **1976–1995**.
- Size: **10,960** firms and **81,095** firm-year observations (Table 1).

2.2 Why state of incorporation matters (Table 1)

BC laws are state laws. The treatment is therefore linked to a firm’s *state of incorporation*, not its location. Table 1 documents that:

- only about **38%** of firms are incorporated in their state of location,
- about **51%** are incorporated in Delaware,
- the remaining firms are incorporated in other states.

Hence, using location would misclassify treatment status for many firms.

3 Measuring competition

3.1 Herfindahl–Hirschman index (HHI): definition (Equation (1))

The paper’s benchmark competition measure is the HHI at the **three-digit SIC** industry-year level:

$$HHI_{jt} := \sum_{i=1}^{N_j} s_{ijt}^2, \quad (1)$$

where s_{ijt} is firm i ’s market share (computed from Compustat sales) within industry j in year t .

Interpretation.

- **Higher HHI** $\Rightarrow$ more concentration $\Rightarrow$ **less competition**.
- **Lower HHI** $\Rightarrow$ more fragmented market $\Rightarrow$ **more competition**.

3.2 Alternative competition measures (Tables 7–8)

To ensure the results are not an artifact of a single measure, the paper also considers:

- HHI based on **two-digit** and **four-digit** SIC codes,
- a **margin-based** measure (industry net profit margin, NPM) as a proxy for market power,
- the **Census HHI** for manufacturing industries (more comprehensive coverage than Compustat),
- **import penetration** for manufacturing, capturing competitive pressure from trade.

4 The governance shock: Business combination laws

4.1 BC laws as a governance instrument

BC laws restrict business combinations between a target and an “interested shareholder” for a specified period (often 3 years) unless certain conditions are met. The empirical interpretation is:

- passage of a BC law $\Rightarrow$ hostile takeovers become more costly / less likely,
- takeover threat weakens $\Rightarrow$ external governance declines.

4.2 Treatment indicator

Let k index state of incorporation. Define:

$$BC_{kt} = 1\{\text{state } k \text{ has passed a BC law by year } t\}.$$

The variation comes from:

- **within-firm time variation:** before vs after a BC law in its incorporation state;
- **cross-sectional variation:** states adopt at different times.

5 Main empirical model

5.1 Baseline panel specification (Equation (2))

Let i index firms, j industries, k states of incorporation, l states of location, and t years. The paper estimates:

$$y_{ijklt} = \alpha_i + \alpha_t + \beta_1 BC_{kt} + \beta_2 HHI_{jt} + \beta_3 (BC_{kt} \times HHI_{jt}) + \gamma' X_{ijklt} + \varepsilon_{ijklt}, \quad (2)$$

where:

- y_{ijklt} is typically **ROA** (operating income before depreciation divided by assets),
- α_i are firm fixed effects and α_t are year fixed effects,
- X_{ijklt} includes key controls (firm size, size squared, age) and two important nonstandard controls:
 - **Industry-year:** average outcome among other firms in the same industry-year,
 - **State-year:** average outcome among other firms in the same location-state-year,designed to absorb contemporaneous industry and local shocks.
- standard errors are clustered at the state of incorporation level.

5.2 Interpreting the coefficients

Average effect of BC laws. When competition is held fixed, the effect is β_1 (evaluated at $HHI = 0$). Empirically, β_1 is typically small and statistically weak.

How competition changes the effect. The marginal effect of BC laws is:

$$\frac{\partial y}{\partial BC} = \beta_1 + \beta_3 HHI.$$

If $\beta_3 < 0$, then the performance impact of weaker governance becomes *more negative* as markets become *less competitive* (higher HHI). This is exactly the “governance matters more in non-competitive industries” prediction.

Table 1

States of incorporation and states of location.

"BC year" indicates the year in which a business combination (BC) law was passed. "State of location" indicates the state in which a firm's headquarters are located. BC years are from Bertrand and Mullainathan (2003). States of location and states of incorporation are both from Compustat. The sample consists of all Compustat firms except regulated utility firms (SIC 4900–4999). The number of firms in the sample is 10,960, and the sample period is from 1976 to 1995.

State	BC Year	State of incorporation	State of location	Number (percentage) of firms incorporated in:			
		Number of firms	Number of firms	State of location	Delaware	Other states	
Delaware	1988	5,587	39	35 (89.7%)		4 (10.3%)	
California		529	1,711	489 (28.6%)	1,034 (60.4%)	188 (11.0%)	
...							
Nebraska	1988	9	29	8 (27.6%)	18 (62.1%)	3 (10.3%)	
West Virginia		8	19	7 (36.8%)	9 (47.4%)	3 (15.8%)	
Montana		7	13	7 (53.8%)	4 (30.8%)	2 (15.4%)	
Vermont		7	16	6 (37.5%)	9 (56.3%)	1 (6.3%)	
Alaska		6	6	4 (66.7%)	2 (33.3%)	0 (0.0%)	
South Dakota		1990	4	10	4 (40.0%)	5 (50.0%)	1 (10.0%)
North Dakota			2	4	1 (25.0%)	2 (50.0%)	1 (25.0%)
Total		10,960	10,960	4,144 (37.8%)	5,552 (50.7%)	1,264 (11.5%)	

Table 2

Summary statistics.

Panel A: ROA (trimmed at 1% level)			
Mean	Median	Minimum	Maximum
0.074	0.104	−1.051	0.417
Panel B: "Eventually BC" states vs. "Never BC" states			
	[1] All states	[2] Eventually BC	[3] Never BC
Size	4.450 (2.283)	4.585 (2.270)	3.629 (2.185)
Age	2.252 (0.918)	2.293 (0.924)	2.002 (0.837)
HHI	0.225 (0.155)	0.226 (0.156)	0.214 (0.148)

5.3 Categorical version (HHI terciles)

The paper also replaces the continuous interaction with tercile interactions:

$$BC \times HHI(Low), \quad BC \times HHI(Medium), \quad BC \times HHI(High),$$

which provides a nonparametric check that effects are monotone across competition bins.

6 Interpreting every table

6.1 Table 1: States of incorporation and location

Purpose: Document incorporation vs location patterns and the timing of BC laws.

Main lessons:

- Delaware dominates as incorporation state (about half of firms).
- Many firms are incorporated outside their headquarters state, so incorporation is essential for correct treatment assignment.
- BC law adoption is staggered across states, creating quasi-experimental time variation.

Table 3

Does corporate governance matter in competitive industries?

BC is a dummy variable that equals one if the firm is incorporated in a state that has passed a BC law. $HHI(Low)$, $HHI(Medium)$, and $HHI(High)$ are dummy variables that equal one if the HHI lies in the bottom, medium, and top tercile, respectively, of its empirical distribution. "Industry-year" and "State-year" are variables that indicate the mean of the dependent variable in the firm's industry and state of location, respectively, excluding the firm itself. $BC\ Year(-1)$ is a dummy variable that equals one if the firm is incorporated in a state that will pass a BC law in one year from now. $BC\ Year(0)$ is a dummy variable that equals one if the firm is incorporated in a state that passes a BC law this year. $BC\ Year(1)$ and $BC\ Year(2+)$ are dummy variables that equal one if the firm is incorporated in a state that passed a BC law one year and two or more years ago, respectively. All other variables are defined in Table 2. The coefficients are estimated using ordinary least squares (OLS). Standard errors are clustered at the state of incorporation level. The sample period is from 1976 to 1995. t-Statistics are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Panel A: Main results				Panel B: Reverse causality	
Dependent variable:	[1] ROA	[2] ROA	[3] ROA	Dependent variable:	ROA
BC	-0.006** (2.25)	0.001 (0.35)		BC Year(-1)	-0.001 (0.17)
BC × HHI		0.033*** (4.95)		BC Year(0)	-0.002 (0.39)
BC × HHI(Low)			0.002 (0.68)	BC Year(1)	-0.000 (0.07)
BC × HHI(Medium)			-0.008** (2.56)	BC Year(2+)	0.004 (0.74)
BC × HHI(High)			-0.012*** (4.59)	BC Year(-1) × HHI	0.001 (0.07)
Industry-year	0.206*** (9.67)	0.206*** (9.60)	0.206*** (9.61)	BC Year(0) × HHI	-0.027** (2.06)
State-year	0.249*** (8.86)	0.249*** (8.83)	0.248*** (8.77)	BC Year(1) × HHI	-0.032*** (4.33)
Size	0.096*** (20.27)	0.097*** (20.38)	0.097*** (20.34)	BC Year(2+) × HHI	-0.034*** (4.15)
Size-squared	-0.007*** (20.09)	-0.007*** (20.42)	-0.007*** (20.53)	Industry-year	0.210*** (7.70)
Age	-0.021*** (5.34)	-0.021*** (5.44)	-0.021*** (5.37)	State-year	0.256*** (7.74)
HHI	0.015* (1.66)	0.025*** (2.58)		Size	0.097*** (20.37)
HHI(Medium)			0.006* (1.88)	Size-squared	-0.007*** (20.44)
HHI(High)			0.008** (2.12)	Age	-0.020*** (5.44)
Firm fixed effects	Yes	Yes	Yes	HHI	0.025** (2.53)
Year fixed effects	Yes	Yes	Yes	Firm fixed effects	Yes
Observations	77,460	77,460	77,460	Year fixed effects	Yes
Adj. R-squared	0.68	0.68	0.68	Observations	77,460
				Adj. R-squared	0.68

6.2 Table 2: Summary statistics (state groups)

Purpose: Compare basic firm characteristics across states that eventually adopt BC laws vs those that never do.

Reading:

- Mean/median firm **size** and **age** are broadly similar across the groups (though not identical).
- Mean HHI is similar across the groups, suggesting that treatment is not mechanically concentrated in either high- or low-competition industries at baseline.

6.3 Table 3: Main results (ROA) and reverse-causality timing

Panel A: main results.

- The interaction term $BC \times HHI$ is **negative and strongly significant**.
- When using HHI terciles, the BC effect is close to zero in **low-HHI (competitive)** industries and becomes **increasingly negative** in medium and high-HHI industries.

Interpretation. BC laws weaken takeover pressure for all firms, but the performance effect appears only where competition is weak enough to allow managerial slack.

Panel B: reverse causality (lead/lag).

- The interaction is *not* negative in the year *before* passage (no evidence of pre-trends consistent with reverse causality).

Table 4

Change in firms' earnings management?

$BC(-12m \text{ to } -6m)$ is a dummy variable that equals one if the firm is incorporated in a BC state and the firm's fiscal year end lies between 12 months and six months prior to the month of the law's passage. $BC(-6m \text{ to } 0m)$, $BC(0m \text{ to } 6m)$, $BC(6m \text{ to } 12m)$, and $BC(12m+)$ are defined analogously. Discretionary accruals are computed as in Dechow, Sloan, and Sweeney (1995). Discretionary current accruals are computed as in Teoh, Welch, and Wong (1998). All other variables are defined in Tables 2 and 3. The coefficients are estimated using OLS. Standard errors are clustered at the state of incorporation level. The sample period is from 1976 to 1995. *t*-statistics are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Panel A: Fiscal year ends		Panel B: Earnings management		
Dependent variable:	ROA	Dependent variable:	[1] Discretionary accruals	[2] Discretionary current accruals
$BC(-12m \text{ to } -6m)$	-0.001 (0.23)	BC	-0.000 (0.12)	0.000 (0.15)
$BC(-6m \text{ to } 0m)$	0.002 (0.51)	$BC \times HHI$	-0.001 (0.28)	-0.003 (0.39)
$BC(0m \text{ to } 6m)$	-0.003 (0.70)	Industry-year	0.375*** (13.97)	0.403*** (21.94)
$BC(6m \text{ to } 12m)$	0.000 (0.04)	State-year	0.007 (0.99)	0.054** (2.52)
$BC(12m+)$	0.003 (0.79)	Size	-0.012*** (5.50)	-0.016*** (10.65)
$BC(-12m \text{ to } -6m) \times HHI$	0.001 (0.08)	Size-squared	0.000 (1.40)	0.001*** (4.86)
$BC(-6m \text{ to } 0m) \times HHI$	-0.006 (0.39)	Age	-0.038*** (17.33)	-0.030*** (13.90)
$BC(0m \text{ to } 6m) \times HHI$	-0.019 (0.81)	HHI	-0.004 (0.55)	-0.004 (0.76)
$BC(6m \text{ to } 12m) \times HHI$	-0.031*** (2.62)	Firm fixed effects	Yes	Yes
$BC(12m+) \times HHI$	-0.036** (4.45)	Year fixed effects	Yes	Yes
Industry-year	0.207*** (9.61)	Observations	63,749	64,070
State-year	0.250*** (8.95)	Adj. <i>R</i> -squared	0.29	0.30
Size	0.097*** (20.38)			
Size-squared	-0.007*** (20.43)			
Age	-0.021*** (5.42)			
HHI	0.025*** (2.59)			
Firm fixed effects	Yes			
Year fixed effects	Yes			
Observations	77,460			
Adj. <i>R</i> -squared	0.68			

- The interaction becomes negative in the passage year and remains negative in subsequent years.

This strengthens the interpretation that the law shock precedes the performance decline (rather than performance predicting law adoption).

6.4 Table 4: Accounting timing and earnings management

Motivation: ROA is an accounting measure; a decline could be an artifact of fiscal-year timing or changing accounting choices.

Panel A (fiscal year ends around the law passage month). The negative interaction appears only in fiscal years that are plausibly *after* exposure to the law (e.g., fiscal years ending 6–12 months after passage and beyond), not before. This supports a real post-law effect rather than a mechanical timing artifact.

Panel B (discretionary accruals). Using standard accrual-based earnings management measures (discretionary accruals and discretionary current accruals), the coefficients on BC and $BC \times HHI$ are small and statistically insignificant. Interpretation: the ROA decline in non-competitive industries is unlikely to be driven by systematic earnings manipulation.

6.5 Table 5: Asset risk / asset mix changes

Motivation: Could ROA fall because firms change their asset mix toward safer/lower-return projects after takeover protection?

Test: The table uses (i) cash-flow volatility and (ii) an asset-beta measure as dependent variables.

Finding: Both BC and $BC \times HHI$ are economically small and statistically insignificant. Interpretation: the evidence is inconsistent with a major post-law shift toward systematically safer assets as the main driver of ROA changes.

6.6 Table 6: Did BC laws reduce takeover threat?

Design: A multiperiod logit (discrete-time hazard) model where the dependent variable is whether the firm is acquired in the next calendar year (data from SDC).

Main message:

- On average, BC laws do not significantly reduce acquisition likelihood across all firms.
- When interacting with competition (HHI terciles), BC laws reduce takeover likelihood **in the most competitive industries** (low HHI) but not in less competitive industries.

Interpretation: This is consistent with the idea that in non-competitive industries the baseline takeover threat may already be low (few potential acquirers), so BC laws have limited incremental effect on takeovers. Yet, even when BC laws reduce takeovers in competitive industries, performance does not deteriorate there—supporting the “competition substitutes for governance” conclusion.

Table 5
Change in firms' asset mix?
Cash-flow volatility is computed as in Zhang (2006). The asset beta is computed as the equity beta times the market value of equity (Compustat item #24 times item #25) divided by the market value of assets (item #24 times item #25 – item #60 + item #6). The equity beta is obtained by estimating the market model over the previous five years using monthly return data from CRSP. All other variables are defined in Tables 2 and 3. The coefficients are estimated using OLS. Standard errors are clustered at the state of incorporation level. The sample period is from 1976 to 1995. t -Statistics are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Dependent variable:	[1] Cash-flow volatility	[2] Asset beta
BC	0.001 (0.53)	0.003 (0.15)
BC $\times$ HHI	0.003 (0.61)	-0.007 (0.30)
Industry-year	0.056*** (4.32)	0.234*** (18.37)
State-year	0.046*** (2.08)	0.215*** (6.65)
Size	-0.033*** (13.57)	0.056*** (4.38)
Size-squared	0.001*** (6.16)	-0.001 (0.95)
Age	0.005*** (2.87)	-0.058** (2.38)
HHI	-0.010 (1.53)	0.109 (1.56)
Firm fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Observations	54,460	75,831
Adj. R-squared	0.63	0.46

Table 6
Did BC laws reduce the takeover threat?
“Likelihood of being acquired” is a dummy variable that equals one if the firm is acquired in the next calendar year. The acquisition data are from the Securities Data Corporation's (SDC) database. All other variables are defined in Tables 2 and 3. The coefficients are estimated using a multiperiod logit model. Standard errors are clustered at the state of incorporation level. The sample period is from 1978 to 1995. z -Statistics are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Dependent variable: Likelihood of being acquired	[1]	[2]
BC	-0.182 (1.59)	
BC $\times$ HHI(Low)		-0.318 * (1.82)
BC $\times$ HHI(Medium)		-0.114 (0.86)
BC $\times$ HHI(High)		-0.029 (0.23)
Industry-year	3.335 *** (7.35)	3.347 *** (7.20)
State-year	2.189 * (1.82)	2.170 * (1.83)
Size	-0.054 ** (2.20)	-0.054 ** (2.22)
Size-squared	0.008 *** (3.26)	0.008 *** (3.29)
HHI(Medium)	-0.060 (0.63)	-0.196 (1.06)
HHI(High)	-0.060 (0.61)	-0.251 (1.20)
Age fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Observations	77,142	77,142
Adj. R-squared	0.06	0.06

Table 7
Alternative measures of competition (all industries).
HHI (2-digit) and HHI (4-digit) are HHIs based on two-digit and four-digit SIC codes, respectively. NPM is operating income before depreciation and amortization (Compustat item #13) divided by sales (item #12). Industry NPM is the median NPM in a given year and three-digit SIC industry. All other variables are defined in Tables 2 and 3. The coefficients are estimated using OLS. Standard errors are clustered at the state of incorporation level. The sample period is from 1976 to 1995. t -Statistics are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Dependent variable: ROA	[1] HHI (2-digit)	[2] HHI (4-digit)	[3] Industry NPM
BC	-0.000 (0.15)	0.000 (0.11)	0.000 (0.07)
BC $\times$ HHI (2-digit)	-0.056 *** (5.15)		
BC $\times$ HHI (4-digit)		-0.022 *** (3.23)	
BC $\times$ Industry NPM			-0.054 *** (3.03)
Industry-year	0.203 *** (9.90)	0.201 *** (9.72)	0.136 *** (9.67)
State-year	0.251 *** (8.76)	0.249 *** (9.26)	0.255 *** (10.98)
Size	0.096 *** (19.30)	0.096 *** (21.35)	0.089 *** (19.40)
Size-squared	-0.007 *** (18.57)	-0.007 *** (21.25)	-0.006 *** (17.98)
Age	-0.021 *** (5.21)	-0.020 *** (4.99)	-0.020 *** (6.26)
HHI (2-digit)	0.011 (0.76)		
HHI (4-digit)		0.017 ** (2.13)	
Industry NPM			0.098 *** (4.58)
Firm fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Observations	77,135	77,446	76,365
Adj. R-squared	0.68	0.68	0.68

Table 8

Alternative measures of competition (manufacturing industries).

HHI (Census) is the HHI based on four-digit SIC manufacturing industries (SIC 2000–3999) provided by the U.S. Census Bureau. The index is available for the years 1982, 1987, and 1992 during the sample period. To fill in the missing years, we always use the index value from the latest available year. For the years prior to 1982, we use the index value from 1982. “Import penetration” is a dummy variable that equals one if the import penetration in a given four-digit SIC manufacturing industry lies above the industry mean. Import penetration is defined as imports divided by the sum of total shipments minus exports plus imports. The import data are from Peter Schott’s Web page and are described in Feenstra (1996) and Feenstra, Romalis, and Schott (2002). All other variables are defined in Tables 2 and 3. The coefficients are estimated using OLS. Standard errors are clustered at the state of incorporation level. The sample period is from 1976 to 1995. *t*-Statistics are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Dependent variable: ROA	[1] <i>HHI</i> (Census)	[2] Import penetration	[3] <i>HHI</i> (Census) & import penetration
<i>BC</i>	−0.003 (0.83)	−0.004 (0.95)	−0.000 (0.09)
<i>BC</i> × <i>HHI</i> (Compustat)			
<i>BC</i> × <i>HHI</i> (Census)	−0.081*** (2.84)		−0.104*** (2.62)
<i>BC</i> × (1 − Import penetration)		−0.007* (1.90)	−0.007 (1.29)
Industry-year	0.148*** (6.21)	0.177*** (8.07)	0.154*** (6.08)
State-year	0.284*** (3.99)	0.348*** (5.87)	0.273*** (2.60)
Size	0.115*** (13.13)	0.097*** (18.57)	0.091*** (13.77)
Size-squared	−0.009*** (12.45)	−0.007*** (17.55)	−0.007*** (13.96)
Age	−0.043*** (5.39)	−0.031*** (5.12)	−0.037*** (5.01)
1-Import penetration		0.011*** (3.09)	0.011** (2.44)
Firm fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Observations	19,244	21,031	17,551
Adj. <i>R</i> -squared	0.73	0.69	0.71

6.7 Table 7: Alternative measures of competition (all industries)

Purpose: Show that the main interaction result is robust to different competition proxies.

Results:

- Using 2-digit HHI, 4-digit HHI, or industry net profit margin (NPM), the interaction term remains economically meaningful and statistically significant.
- The sign pattern is consistent: weaker competition (higher concentration / higher margins) amplifies the negative effect of BC laws on ROA.

6.8 Table 8: Alternative measures of competition (manufacturing industries)

Purpose: Use competition measures available only for manufacturing:

- Census HHI (broader firm coverage than Compustat),
- import penetration (competitive pressure from trade).

Results:

- The interaction $BC \times HHI(\text{Census})$ is negative and significant.
- The interaction with (low) import penetration is consistent with stronger negative effects where foreign competition is weaker.

6.9 Table 9: Mechanisms — empire building vs quiet life

This table asks *what agency problem competition mitigates*.

Table 9

Empire building or quiet life?

In Panel A, capital expenditures (Compustat item #30) are divided by total assets (item #6). Asset growth is the percentage increase in total assets from one year to the next. PPE growth is the percentage increase in property, plant, and equipment (item #8) from one year to the next. "Acquisition ratio" is the sum of the value of all acquisitions made by the firm in a given year divided by the firm's average market capitalization in that year (from CRSP). The acquisition data are from the Securities Data Corporation's (SDC) database. "Likelihood of being acquirer" is a dummy variable that equals one if the firm makes at least one acquisition during the year. In Panel B, Selling, general & admin. expenses are SG&A expenses (item #189) divided by total assets. Advertising expenses (item #45) and Costs of goods sold (item #41) are both divided by sales (item #12). R&D expenses (item #46) are divided by total assets. Wages (real) are the natural logarithm of labor and related expenses (item #42) divided by the number of employees (item #29) and deflated by the consumer price index from the U.S. Bureau of Labor Statistics. All other variables are defined in Tables 2 and 3. For brevity, only the coefficients on BC and $BC \times HHI$ are reported. The coefficients are estimated using OLS. Standard errors are clustered at the state of incorporation level. The sample period is from 1976 to 1995, except in columns 4 and 5 of Panel A, where the sample period is from 1979 to 1995. t -Statistics are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Panel A: Empire building					
Dependent variable:	[1] Capital expenditures	[2] Asset growth	[3] PPE growth	[4] Acquisition ratio	[5] Likelihood of being acquirer
BC	-0.000 (0.27)	-0.004 (0.70)	-0.003 (0.53)	0.000 (0.02)	0.003 (0.42)
$BC \times HHI$	0.001 (0.18)	-0.004 (0.39)	-0.000 (0.04)	0.001 (0.43)	-0.010 (0.48)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	74,435	67,806	64,449	70,248	75,415
Adj. R-squared	0.55	0.35	0.22	0.29	0.36
Panel B: Quiet life					
Dependent variable:	[1] Selling, general & admin. expenses	[2] Advertising expenses	[3] R&D expenses	[4] Costs of goods sold	[5] Wages (real)
BC	0.005 (0.80)	0.000 (0.59)	0.001 (0.49)	-0.002 (0.20)	-0.003 (0.12)
$BC \times HHI$	0.029** (2.51)	0.003 (1.04)	0.007 (1.39)	0.053 ** (2.44)	0.103** (2.00)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	68,561	28,389	39,359	74,758	8,651
Adj. R-squared	0.81	0.80	0.76	0.60	0.89

Panel A: empire building. Outcomes include capital expenditures, asset growth, PPE growth, acquisition activity. Neither BC nor $BC \times HHI$ is significant. Interpretation: there is little evidence that takeover protection leads managers in non-competitive industries to expand the firm aggressively.

Panel B: quiet life. Outcomes include SG&A (overhead), advertising, R&D, cost of goods sold (input costs), and real wages. The interaction $BC \times HHI$ is **positive and significant** for overhead costs, input costs, and real wages. Interpretation: when shielded from takeovers *and* weak competition, managers appear to tolerate higher costs and pay higher wages—a “quiet life” channel that mechanically reduces profitability/ROA.

6.10 Table 10: Event-study results

Question: Do stock prices reflect the differential impact of BC laws across competition environments?

Event date: first newspaper report about the BC law (not final passage, which is anticipated).

Method: state portfolios + market model (see Section 7).

Panel A:

- In the two-day event window $[-1, 0]$, the average CAR is negative overall.
- The CAR is close to zero for low-HHI (competitive) portfolios, and **more negative** for high-HHI portfolios, with a monotone pattern across HHI partitions.

Panel B: A long-short hedge portfolio (long top-HHI, short bottom-HHI) earns positive

Table 10

Event-study results.

The methodology used to compute cumulative abnormal returns (CARs) is described in Section 5. The event date is the date of the first newspaper report about the BC law. The two-day event window is denoted by $[-1, 0]$. The numbers reported in the table are average portfolio CARs based on 19 state portfolios. The 19 states are Arizona, Connecticut, Delaware, Georgia, Illinois, Kentucky, Maryland, Massachusetts, Minnesota, New Jersey, New York, Ohio, Oklahoma, Pennsylvania, Tennessee, South Carolina, Virginia, Washington, and Wisconsin. The construction of the state-HHI portfolios, bottom-HHI portfolios, and top-HHI portfolios is described in Section 5. In Panel B, the hedge portfolio is long in the top-HHI portfolio and short in the bottom-HHI portfolio. z-Statistics are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Panel A: Event-study results						
	[1] All firms	[2] HHI (Low)	[3] HHI (High)	[4] HHI (Low)	[5] HHI (Medium)	[6] HHI (High)
$[-40, -2]$	0.98 (1.44)	1.25 (1.40)	0.61 (0.49)	1.51 (1.53)	2.11 (1.13)	-0.30 (0.04)
$[-30, -2]$	0.43 (0.94)	0.83 (1.08)	0.08 (0.07)	0.78 (1.02)	0.52 (0.36)	-0.34 (0.07)
$[-20, -2]$	0.08 (0.53)	0.15 (0.47)	-0.01 (0.22)	0.33 (0.78)	-0.07 (-0.03)	-0.41 (0.15)
$[-10, -2]$	0.52 (1.35)	0.44 (1.31)	0.57 (0.54)	0.44 (1.19)	1.15 (1.24)	0.10 (0.21)
$[-3, -2]$	-0.02 (0.05)	0.22 (0.47)	-0.24 (-0.50)	0.38 (0.75)	0.09 (-0.26)	-0.24 (-0.25)
$[-1, 0]$	-0.32*** (-2.58)	-0.10 (-1.29)	-0.54** (-2.36)	0.08 (-0.53)	-0.44* (-1.67)	-0.67** (-2.31)
$[1, 2]$	0.09 (0.37)	-0.03 (0.07)	0.20 (0.45)	0.01 (-0.05)	0.25 (1.02)	0.03 (-0.28)
$[1, 10]$	-0.07 (-0.08)	0.03 (0.07)	-0.17 (-0.07)	0.30 (0.78)	-0.74 (-0.53)	-0.27 (-0.61)
Panel B: Hedge portfolios						
HHI Partition	Median	Terciles	Quartiles	Quintiles		
Bottom-HHI portfolio	-0.10 (-1.29)	0.08 (-0.53)	0.17 (-0.62)	0.19 (-0.64)		
Top-HHI portfolio	-0.54** (-2.36)	-0.67** (-2.31)	-0.75** (-2.44)	-0.78** (-2.49)		
Hedge portfolio	0.44 (1.41)	0.75* (1.76)	0.92** (2.02)	0.97** (2.06)		

abnormal returns around the event date, and the magnitude grows with finer HHI partitions. Interpretation: the market anticipates that BC laws are most harmful in less competitive industries.

7 Event-study model block

7.1 Market model estimation (Equation (3))

For each incorporation-state portfolio j , estimate the market model using daily returns in an estimation window prior to the event:

$$R_{jt} = a_j + b_j R_{mt} + e_{jt}, \quad (3)$$

where R_{jt} is the daily return of the equally weighted portfolio of firms incorporated in state j , and R_{mt} is the daily return of the equally weighted CRSP market portfolio.

7.2 Abnormal returns and CARs (Equation (4))

Define the normal return as $\hat{R}_{jt} = \hat{a}_j + \hat{b}_j R_{mt}$. Then the abnormal return is:

$$AR_{jt} := R_{jt} - \hat{R}_{jt}. \quad (4)$$

Cumulative abnormal returns (CARs) over a window $[t_1, t_2]$ are:

$$CAR_j[t_1, t_2] = \sum_{t=t_1}^{t_2} AR_{jt}.$$

7.3 Why state portfolios matter

Because the same law affects many firms at once (all firms incorporated in a state), individual-firm event study residuals would be cross-sectionally correlated. Forming state portfolios reduces this dependence and makes inference more reliable.

8 How everything connects

The paper’s logic is internally consistent:

1. BC laws weaken external takeover discipline (the intended governance shock).
2. Product market competition differs across industries (HHI and alternatives).
3. Performance declines *only* when both conditions hold:
 - external governance is weakened, *and*
 - competition is weak enough to allow managerial slack.
4. Mechanism evidence points to a “quiet life” channel (higher overhead and input costs) rather than empire building.
5. Stock prices respond in a way consistent with the real effects being concentrated in non-competitive industries.

9 Conclusion

9.1 Summary

Thus paper study whether the real effects of external corporate governance depend on the level of product-market competition. They exploit the staggered passage of U.S. state *business combination (BC) laws*—anti-takeover statutes that weaken hostile takeover threats—as a plausibly exogenous shock that reduces external governance.

The paper’s central conclusion is that **product-market competition and external governance are substitutes**:

- (i) After BC laws pass, **operating performance declines in non-competitive industries**, while **performance is largely unchanged in competitive industries**. Empirically, this appears as a strongly negative interaction between BC laws and industry concentration (HHI): governance weakening hurts firms *more* when competition is weak.
- (ii) The performance decline is **not driven by accounting manipulation, changes in fiscal-year timing, or a shift to systematically safer assets**. The paper runs targeted checks and finds little evidence that the ROA results reflect earnings management or a risk-reduction story.

- (iii) The evidence favors a “**quiet life**” mechanism rather than **empire building**. When takeover pressure weakens in *non-competitive* industries, managers allow higher costs (e.g., SG&A/overhead, input costs) and higher real wages, consistent with reduced effort to minimize costs. In contrast, there is little evidence of increased investment/asset growth or acquisition-driven expansion that would indicate empire building.
- (iv) Financial markets appear to understand these heterogeneous real effects. Around BC-law news, stock prices respond more negatively for firms in less competitive industries, consistent with investors anticipating that takeover protection is most harmful where competition does not strongly discipline managers.

9.2 Why competition substitutes for external governance

The mechanism can be summarized as “two sources of discipline” that can partially replace each other:

1) External governance discipline (takeover threat). In many settings, the possibility of a hostile takeover disciplines managers: poor performance increases vulnerability to takeover and potential replacement. BC laws reduce that threat, weakening an external disciplining force.

2) Product-market discipline (competition). When competition is intense, firms that allow inefficiency (high costs, slack) lose market share and profitability quickly. This creates an *operating* discipline mechanism: managers must keep the firm efficient to survive, even if takeovers are less likely.

3) Why the effect is stronger in non-competitive industries. In concentrated markets (high HHI), profits are less sensitive to marginal inefficiency, so managers can more easily enjoy a “quiet life” (higher costs, less pressure to optimize). Thus, reducing takeover pressure has a larger marginal impact on behavior and performance when competition is weak.

4) Why the takeover evidence does not overturn the conclusion. The paper finds that BC laws reduce takeover likelihood mainly among firms in *competitive* industries. This is consistent with the idea that takeovers are more feasible when industries are competitive (more potential acquirers, fewer strategic barriers). However, even when takeover probabilities fall in competitive industries, performance does not deteriorate there—precisely because competition already disciplines managerial slack. So the takeaway is not “BC laws do nothing”; it is that *competition can neutralize the real consequences of weakened takeover governance*.

9.3 Takeaway

Weakening takeover governance via business combination laws reduces operating performance primarily in non-competitive industries, consistent with a quiet-life agency problem that competition mitigates, implying that product-market competition and external governance act as substitutes.